COMM 401: Signals & Systems Theory

Lecture 2

- Periodicity

- Famous Signals

Periodic-Complex CT Exponential Signal

- Given the signal:

$$x(t) = e^{j\omega_o t}$$

An important property of this signal is that it is periodic. We know that *x(t)* is periodic with period *T* if *x(t)=x(t+T)*, then:

$$e^{j\omega_o t} = e^{j\omega_o (t+T)} = e^{j\omega_o t} e^{j\omega_o T} \qquad (1)$$

For periodicity, we must have:

$$e^{j\omega_o T} = 1 \qquad (2)$$

Excluding the trivial solution $\omega_o = 0$, then the fundamental period

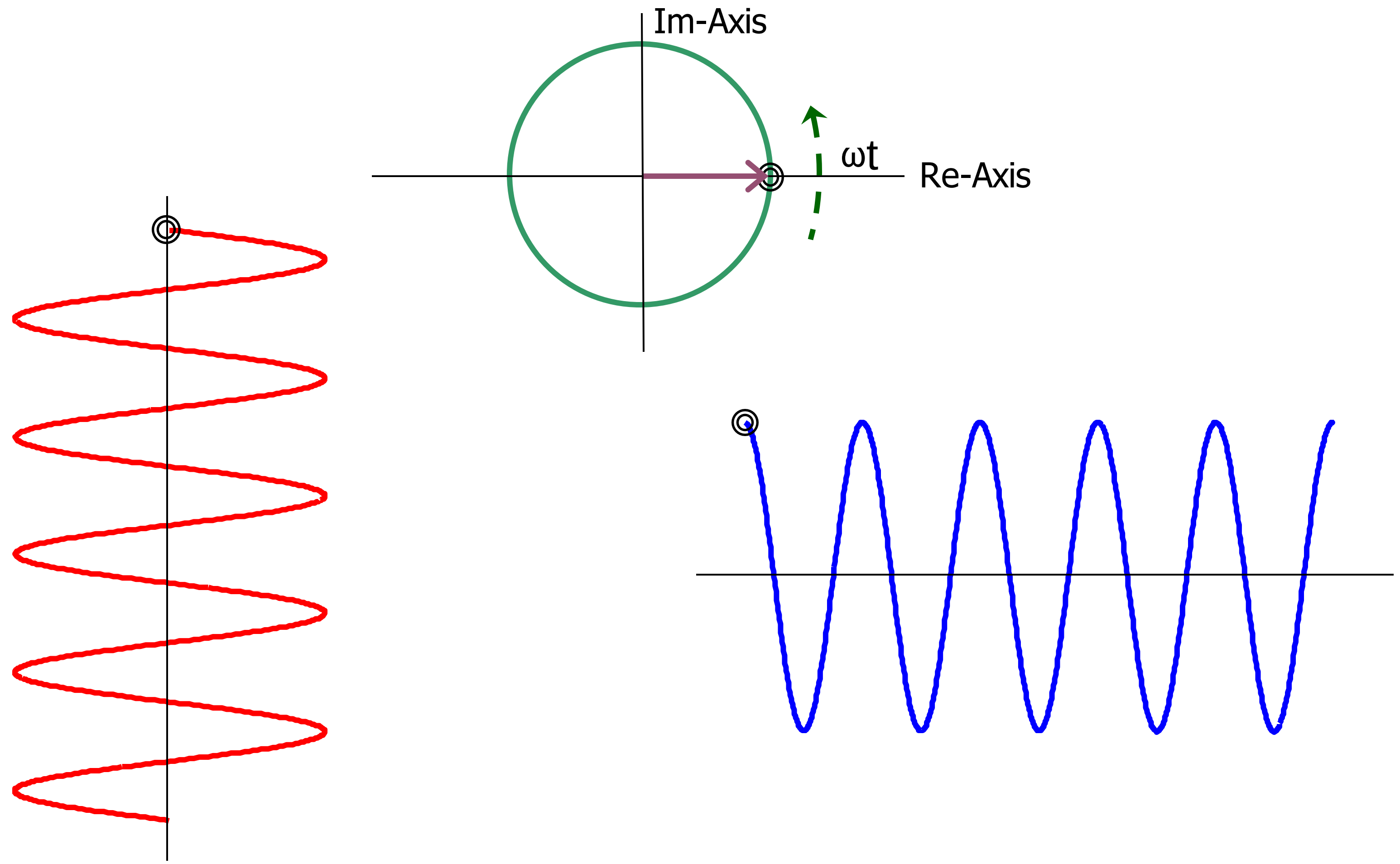
T_o *of* $x(t)$ is the smallest positive value of T for which (2) holds:

$$T_o = \frac{2\pi}{|\omega_o|} \qquad (3)$$

Thus, the signals $e^{j\omega_o t}$ and $e^{-j\omega_o t}$ have the same fundamental period.

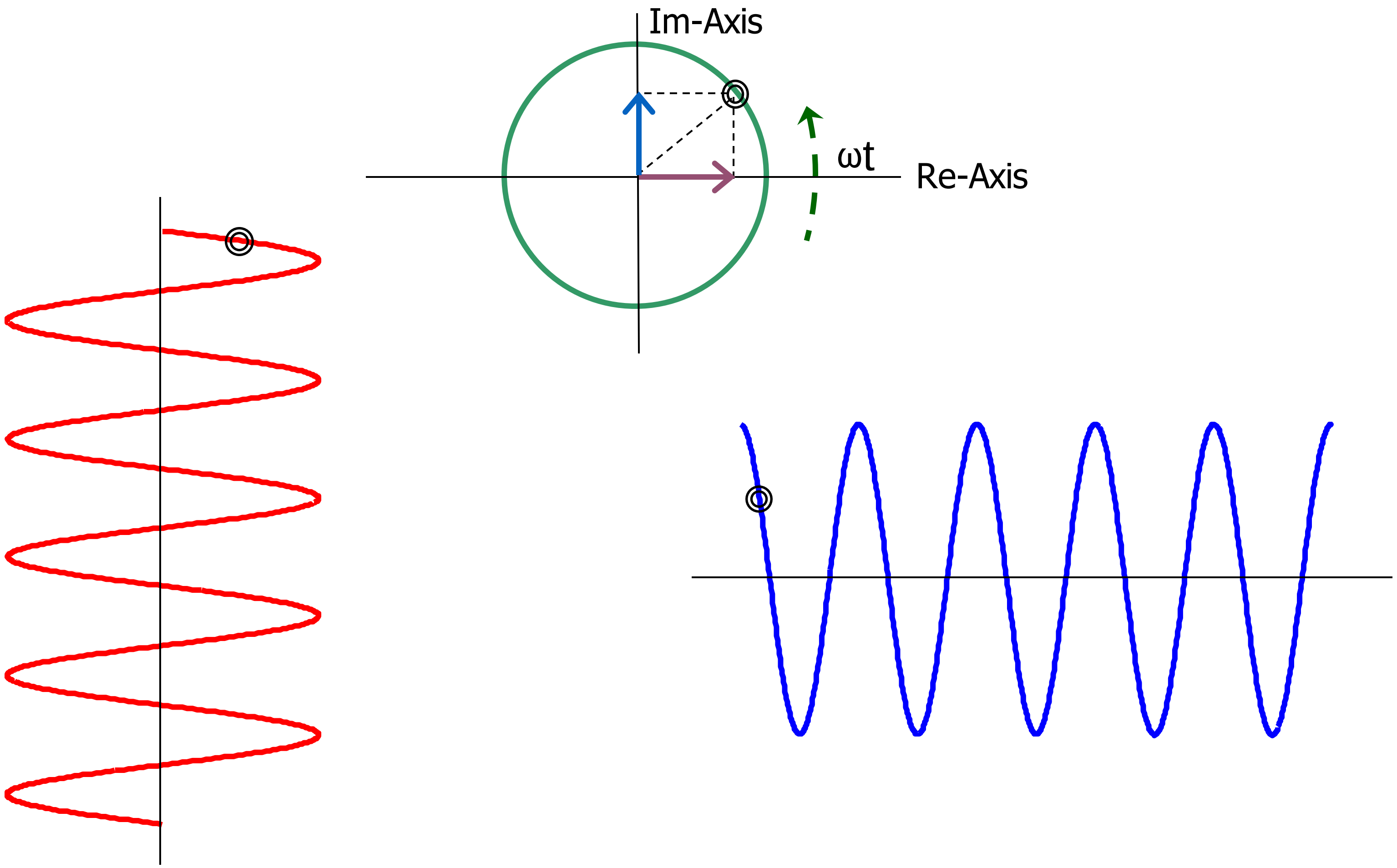
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



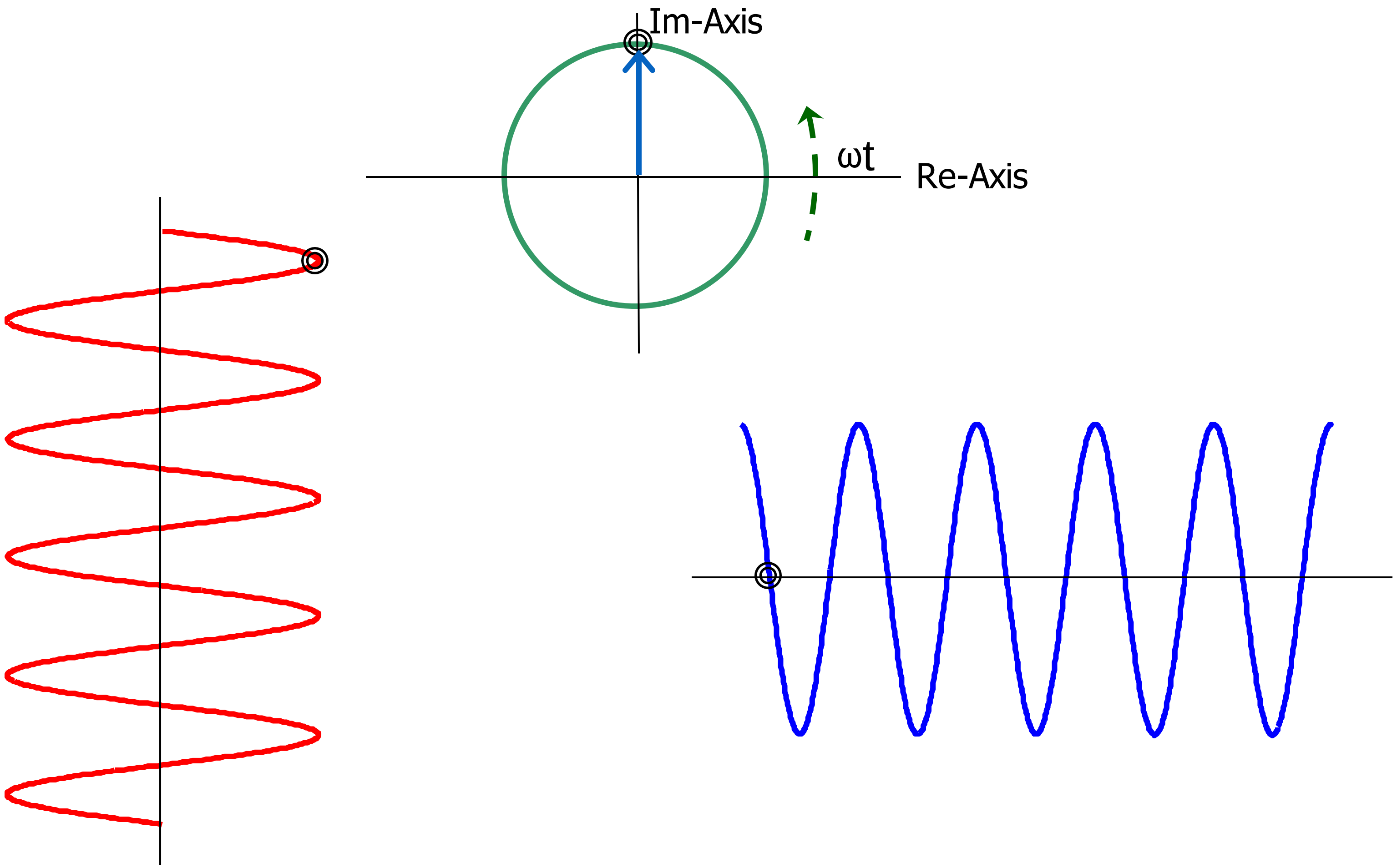
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



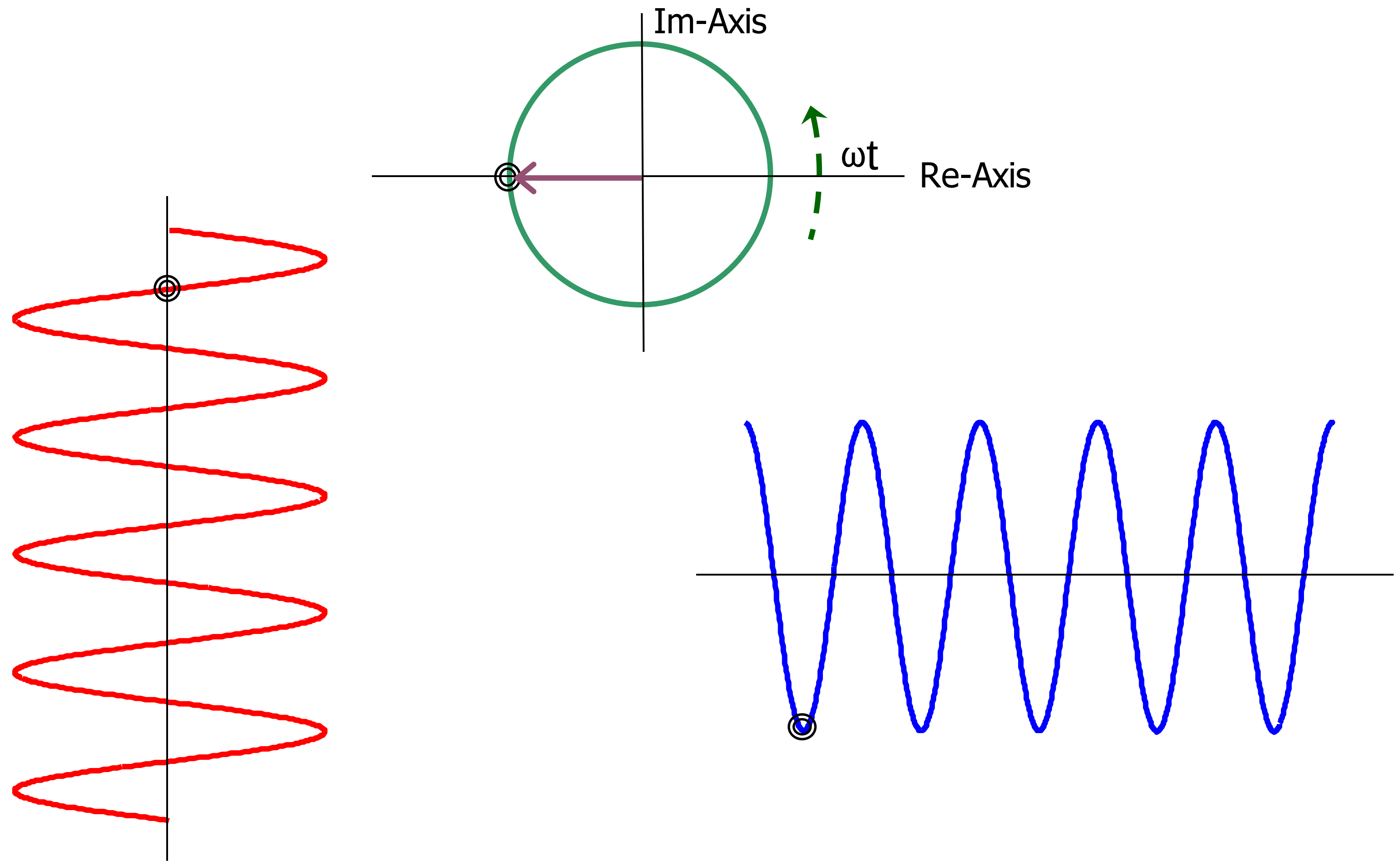
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



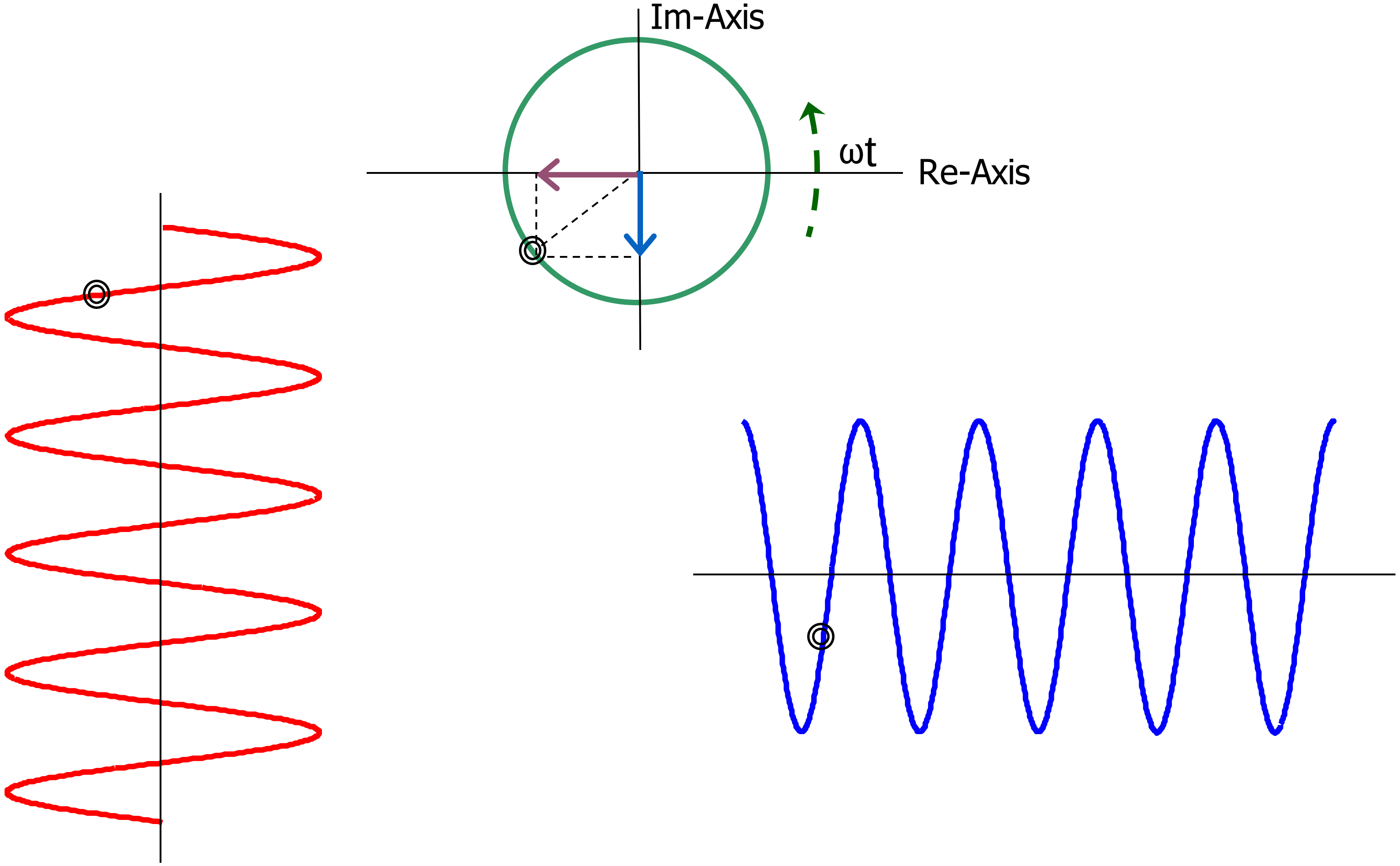
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



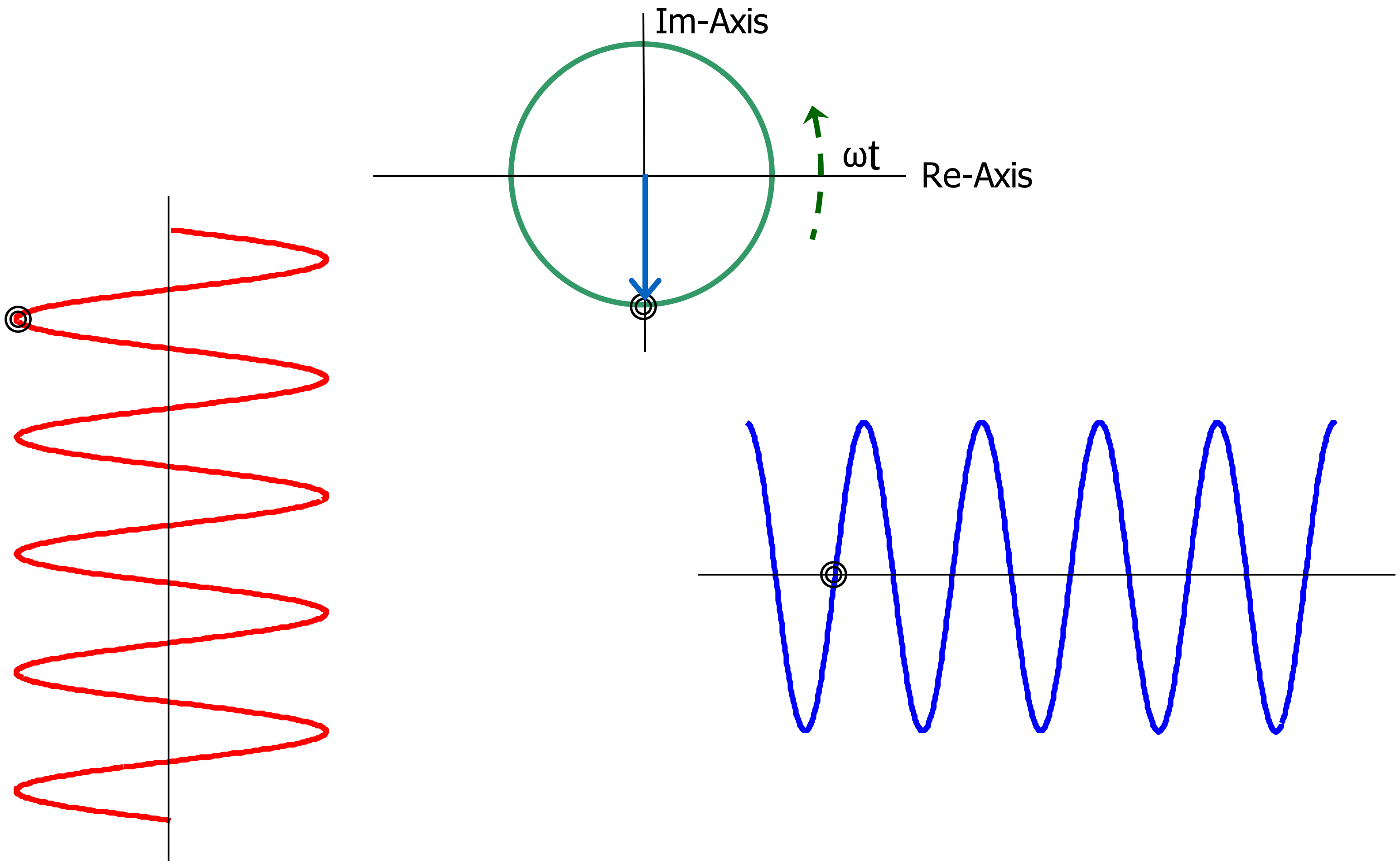
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



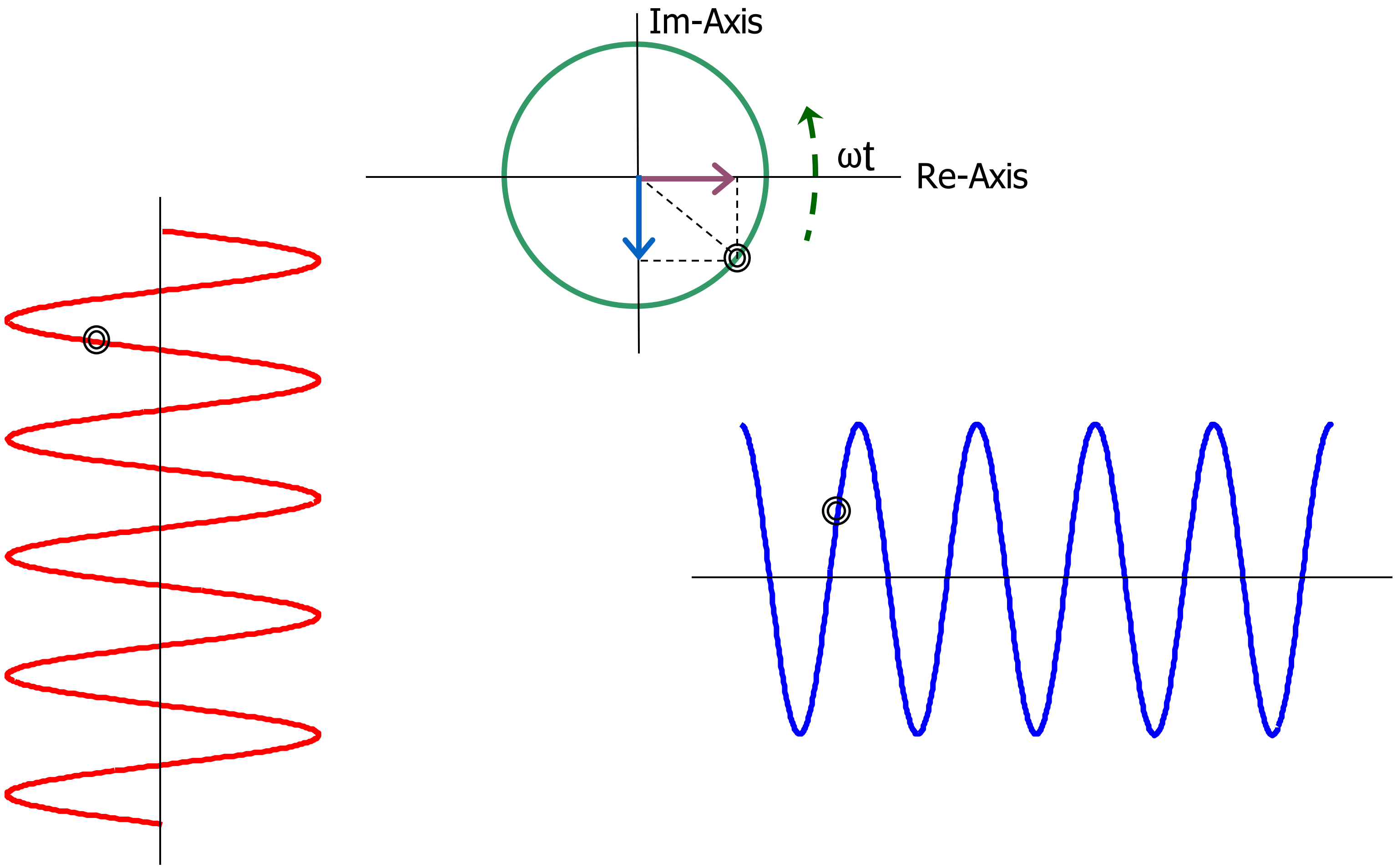
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



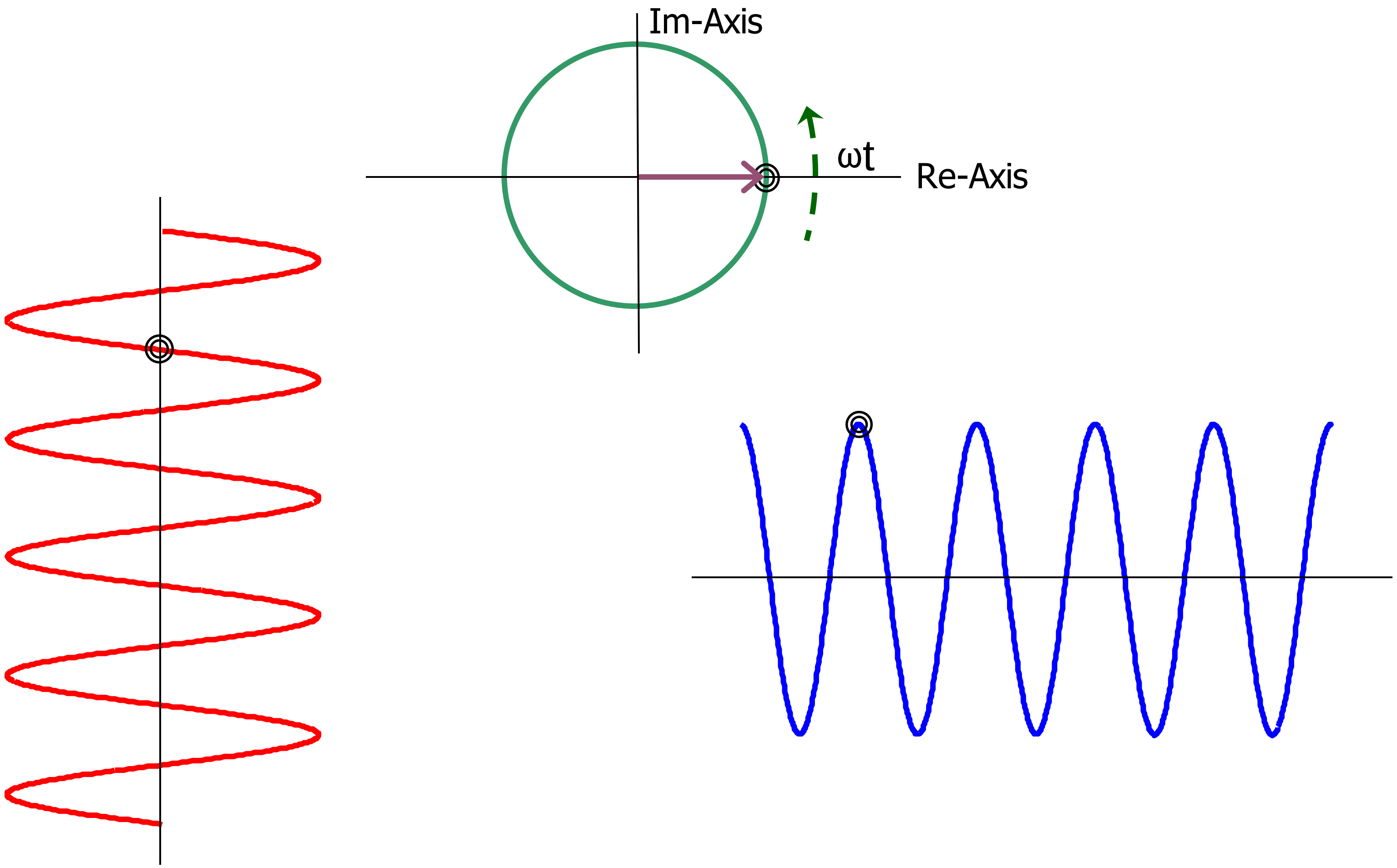
Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$

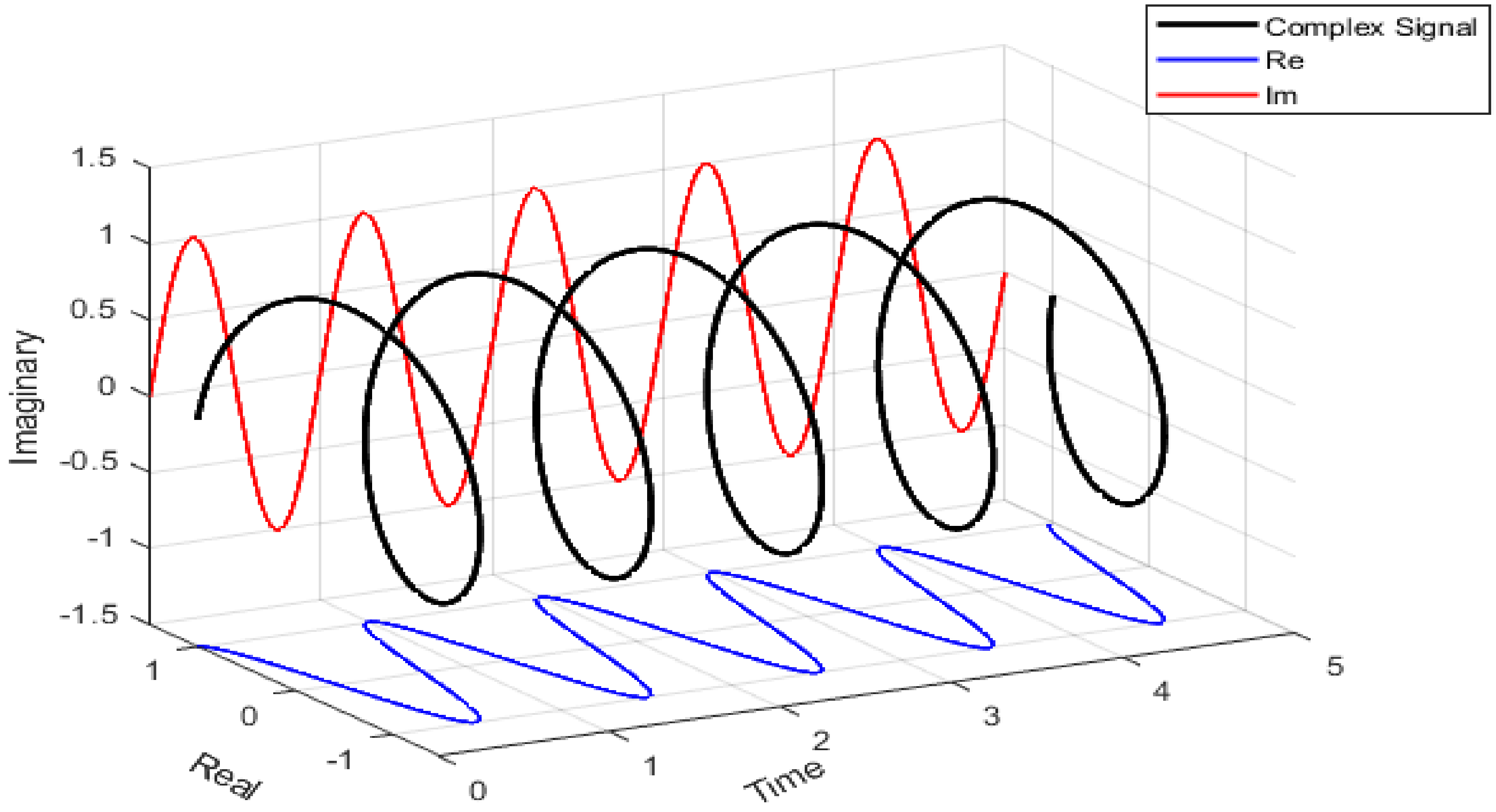


Complex Exponential Function

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$



Complex Exponential Function



Periodicity in Discrete Time

GUC German University in Cairo

Accordingly to the definition of periodicity for period $N > 0$

$$e^{j\omega_0(n+N)} = e^{j\omega_0 n}$$

or equivalently

$$e^{j\omega_0 N} = 1$$

This condition can be fulfilled if there exist such integer number m that

Important !!!!

$$\omega_0 N = 2\pi m \Rightarrow N_0 = m\frac{2\pi}{\omega_0}$$

or equivalently

$$\boxed{\frac{\omega_0}{2\pi} = \frac{m}{N}} \Rightarrow$$

$\frac{m}{N}$ is a rational Number (because m & N are integers)

Conclusion: The signal $e^{j\omega_0 n}$ is periodic only if $\omega_0/(2\pi)$ is a rational number. The fundamental period of such signal is

$$N_0 = m\frac{2\pi}{\omega_0}$$

This is valid also for sinusoidal signals

Examples: Is the following signals are periodic??

$$(a)\ \cos(0.01\pi n); \quad We\ have\ \ \omega_0 = 0.01\pi$$

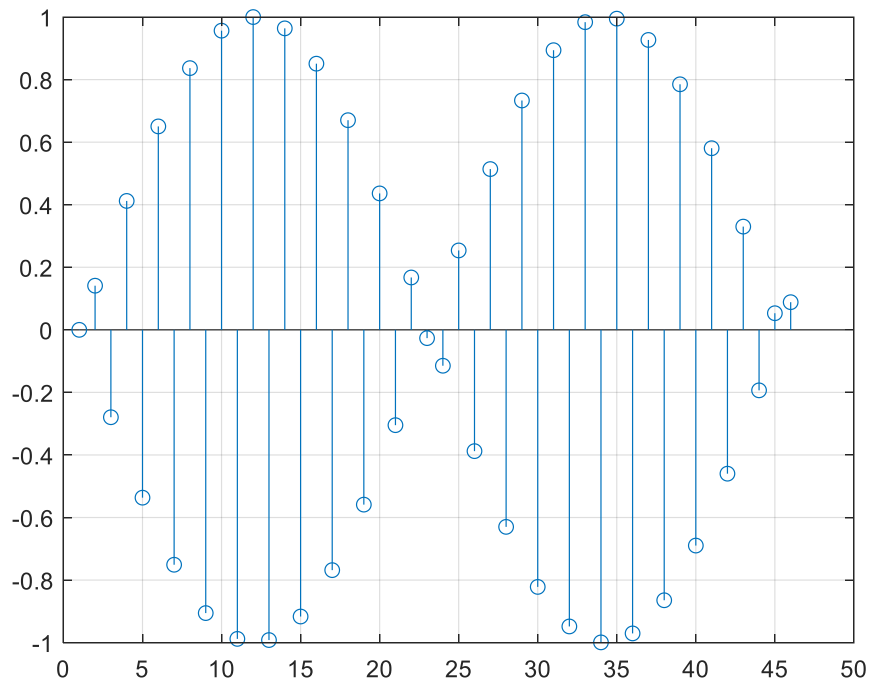
$$then\ \frac{\omega_0}{2\pi} = \frac{0.01\times\pi}{2\pi} = \frac{1}{200}\ which\ is\ rational\ ,\ then$$

$$the\ signal\ is\ periodic\ with\ period\ \ 200$$

$$(b)\sin(3n); \quad We\ have\ \ \omega_0 = 3$$

$$then\ \frac{\omega_0}{2\pi} = \frac{3}{2\pi} = which\ is\ irrational\ ,\ then\ the\ signal\ is$$

$$aperiodic\ (non-periodic).$$

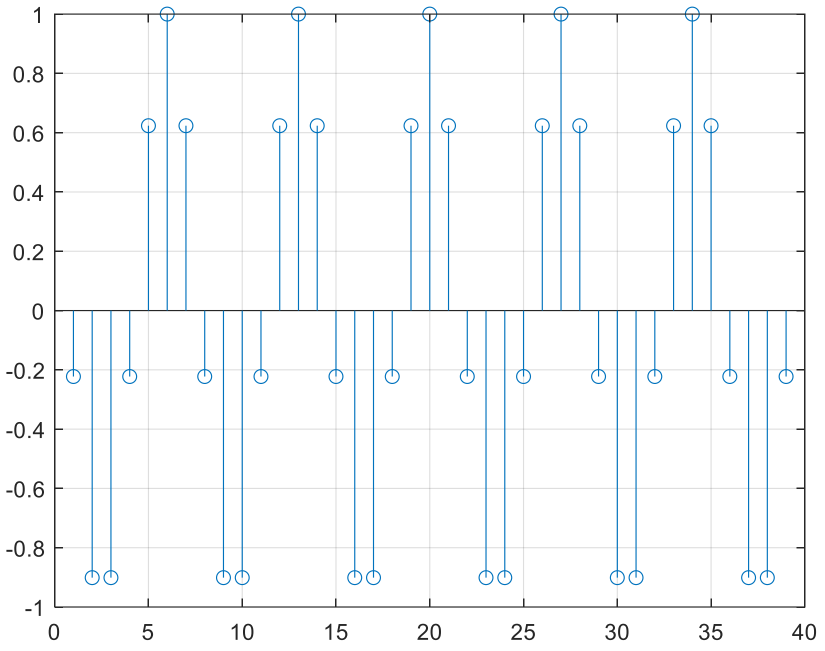


Examples:

$$(c)\cos\left(\frac{30\pi n}{105}\right); \quad We\ have\ \ \omega_0 = \frac{30\pi}{105}$$

$$then\ \frac{\omega_0}{2\pi} = \frac{30\pi}{105\times 2\pi} = \frac{30}{210} = \frac{1}{7}\ which\ is\ rational\ ,\ then\ the\ signal\ is$$

$$periodic\ with\ period\ = N_0 = 7.$$



$$(d)e^{j7\pi n}; We\ have\ \ \omega_0 = 7\pi$$

$$then\ \frac{\omega_0}{2\pi} = \frac{7\pi}{2\pi} = \frac{7}{2}\ which\ is\ rational\ ,\ then\ the\ signal\ is$$

$$periodic\ with\ period\ = N_0 = 2.$$

Difference between Periodicity in CT and DT

CT

$e^{j\omega_0 t}$

$cos(\omega_0 t)$

$sin(\omega_0 t)$

Are periodic for any value of ω_0

DT

$e^{j\omega_0 n}$

$cos(\omega_0 n)$

$sin(\omega_0 n)$

Are periodic if:

$\frac{\omega_0}{2\pi}$ **is a rational number**

Difference between Periodicity in CT and DT

Increase ω_0 by 2π

CT $\quad e^{j\omega_0 t}$

$e^{j2\pi t} \neq 1$

$$e^{j(\omega_0+2\pi)t} = e^{j\omega_0 t} e^{j2\pi t} \neq e^{j\omega_0 t}$$

From the previous equation we see that the exponential at Frequency $(\omega_0+2\pi)$ is different from that one at frequency ω_0

DT $\quad e^{j\omega_0 n}$

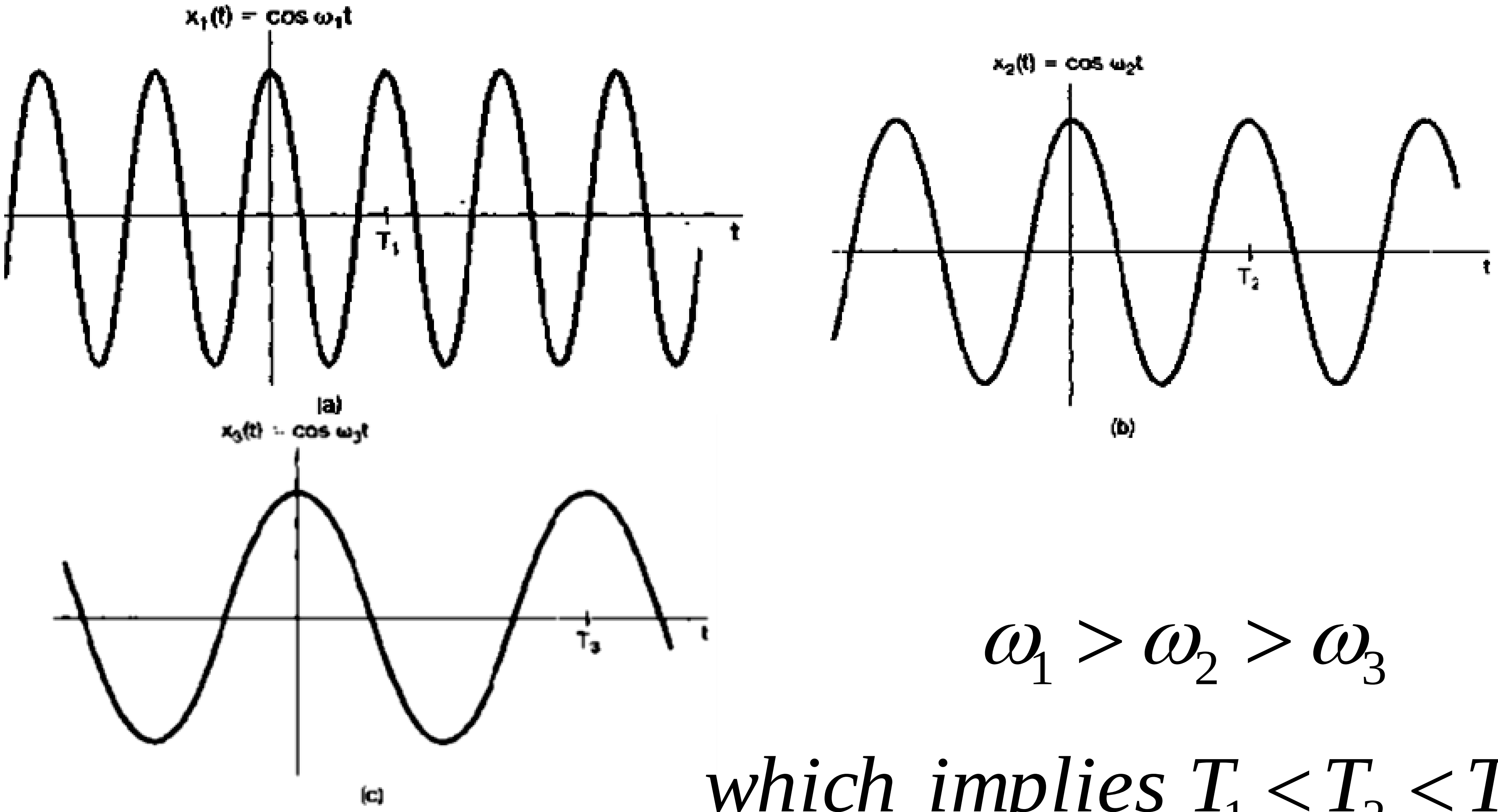
$e^{j2\pi n} = 1$

$$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n} e^{j2\pi n} = e^{j\omega_0 n}$$

From the previous equation we see that the exponential at Frequency $(\omega_0+2\pi)$ is the same as that at frequency ω_0

Relationship Between Fundamental Frequency and Period

Note that the fundamental period is inversely proportional to ω_o as shown in the figure:



$$\omega_1 > \omega_2 > \omega_3$$

$$which\ \ implies\ T_1 < T_2 < T_3$$

Notes in Periodicity in CT and DT

(1) The DC signal is periodic with arbitrary period

(2) periodic $\pm$ non-periodic = non-periodic

(3) periodic $\times$ non-periodic = non-periodic

(4) For DT Signals:

periodic $\pm$ periodic = periodic

(4) For CT Signals:

periodic $\pm$ periodic = periodic $\Rightarrow$ only if $\frac{T_{01}}{T_{02}}$ (or $\frac{\omega_{01}}{\omega_{02}}$) is a rational number.

where: T_{01} is the period of the first signal and T_{02} is the period for the second signal.

The total period T_0 is the least common multiplier of the two periods:

$T_0 = LCM(T_{01}, T_{02})$.

Example:

$$S(t) = \cos(4t)\sin(8t) = \frac{1}{2}[\sin(4t) + \sin(12t)]$$

$$sin(u+v) = \sin(u)\cos(v) + \cos(u)\sin(v)$$

$$\sin(u)\cos(v) = \frac{1}{2}[sin(u+v) + sin(u-v)]$$

$$T_{01} = \frac{2\pi}{4} = \frac{\pi}{2} \qquad T_{02} = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$\frac{T_{01}}{T_{02}} = 3$$

$$\frac{\pi}{2}: \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

$$\frac{\pi}{6}: \frac{\pi}{6}, \frac{2\pi}{6}, \frac{3\pi}{6}, \frac{4\pi}{6}$$

$S(t)$ is periodic with period equal to $T_0 = LCM\left(\frac{\pi}{2}, \frac{\pi}{6}\right) = \frac{\pi}{2}$

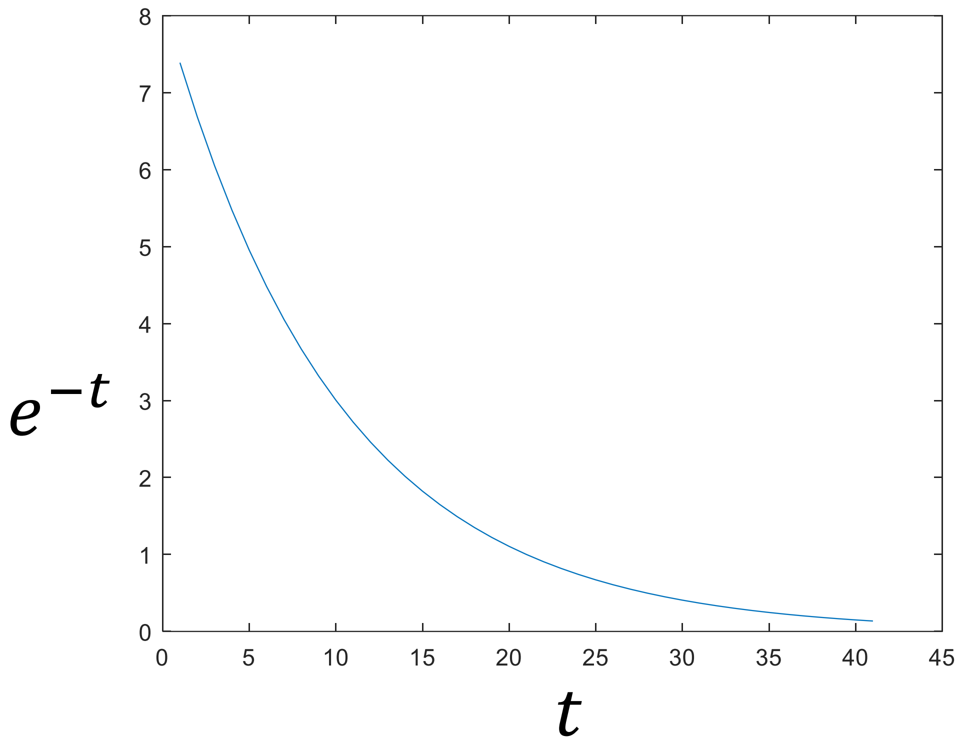
Example:

$$x(t) = e^{(j-1)t}$$

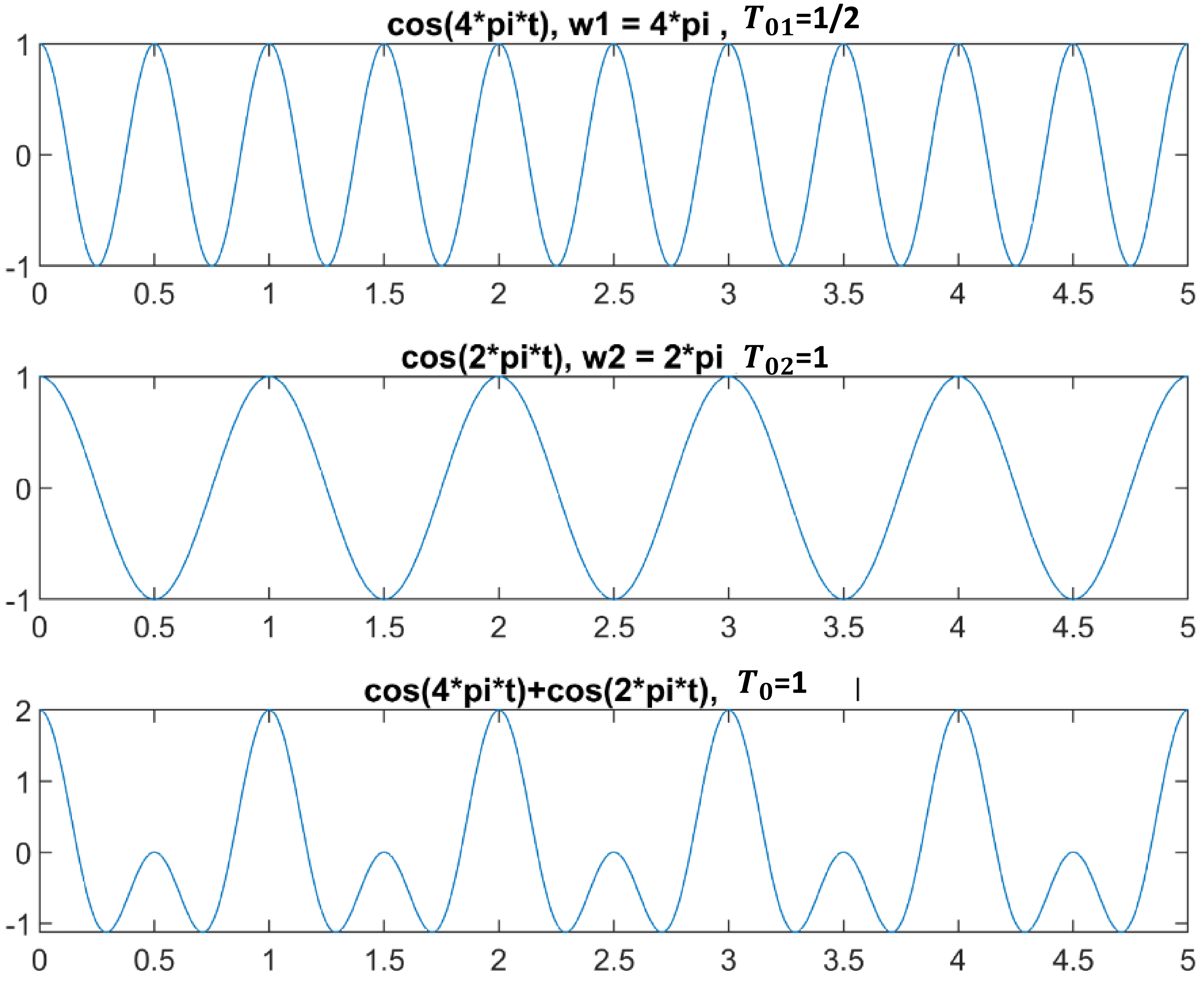
$$= e^{jt} e^{-t}$$

Complex exponential signal.
It is periodic

Real exponential
(non-periodic signal)



so x(t) is non-periodic signal



$$S(t) = \cos(4\pi t) + \cos(2\pi t)$$

$$T_{01} = \frac{2\pi}{4\pi} = \frac{1}{2}$$

$$T_{02} = \frac{2\pi}{2\pi} = 1$$

$\frac{T_{01}}{T_{02}} = \frac{1}{2}$ rational number

$Then\ S(t)\ is\ periodic$

$$T_0 = LCM\left(\frac{1}{2}, 1\right) = 1$$

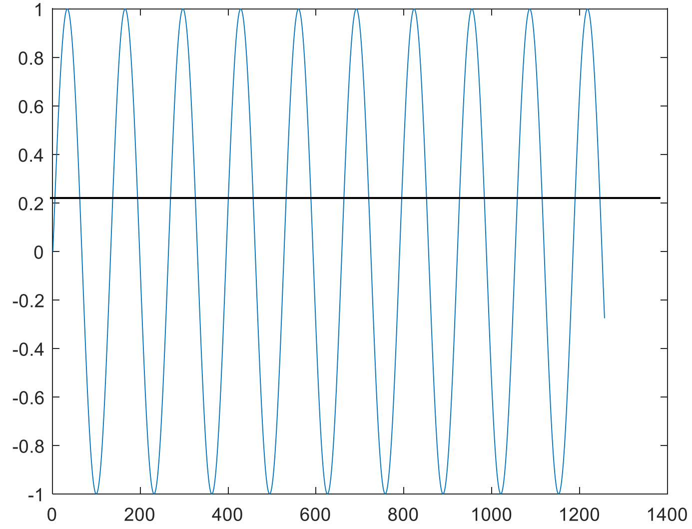
Example: Check periodicity of x(t): $x(t) = \sin(3t) + \sin(4t)$

$$T = \frac{2\pi}{\omega_0} \Rightarrow T_{01} = \frac{2\pi}{3} \qquad T_{02} = \frac{2\pi}{4} = \frac{\pi}{2} \Rightarrow \frac{T_{01}}{T_{02}} = \frac{4}{3}$$ is a rational number

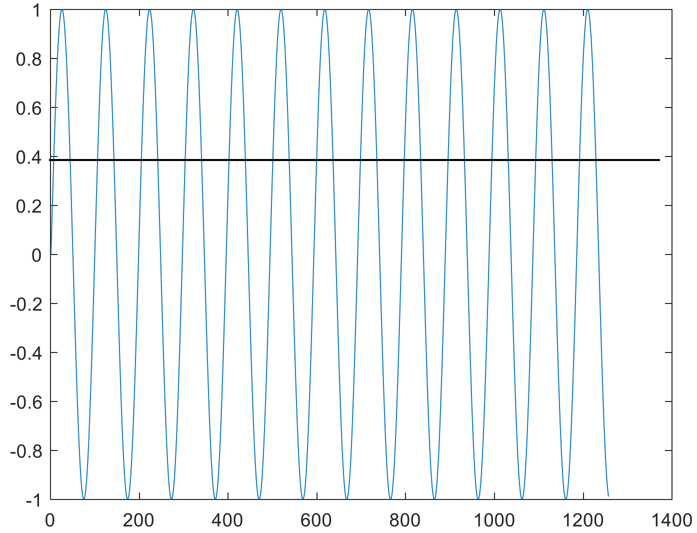
$$T_0 = LCM(T_{01}, T_{02}) = 2\pi$$

$Then\ x(t)$ is periodic with period T_0

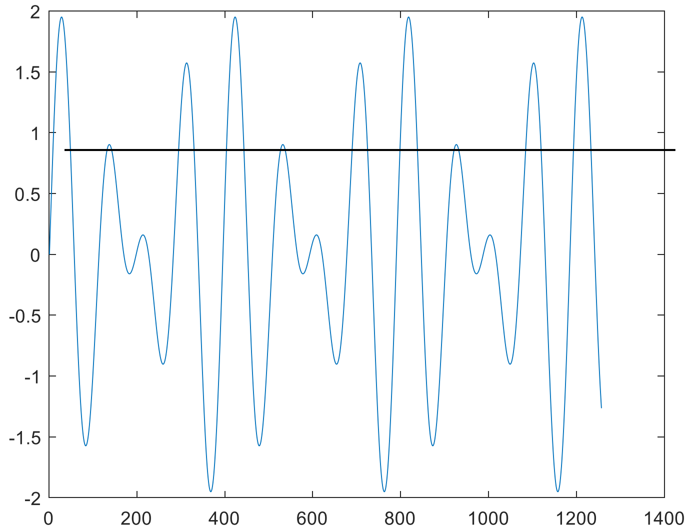
$\sin(3t)$



$\sin(4t)$



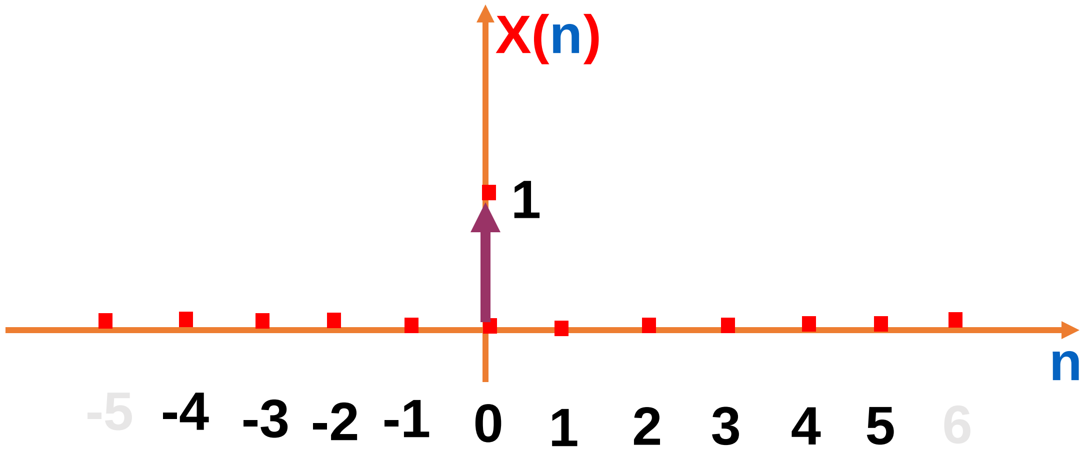
$x(t)$



DT Unit Impulse Sequence

The Unit impulse (unit sample sequence) is described by:

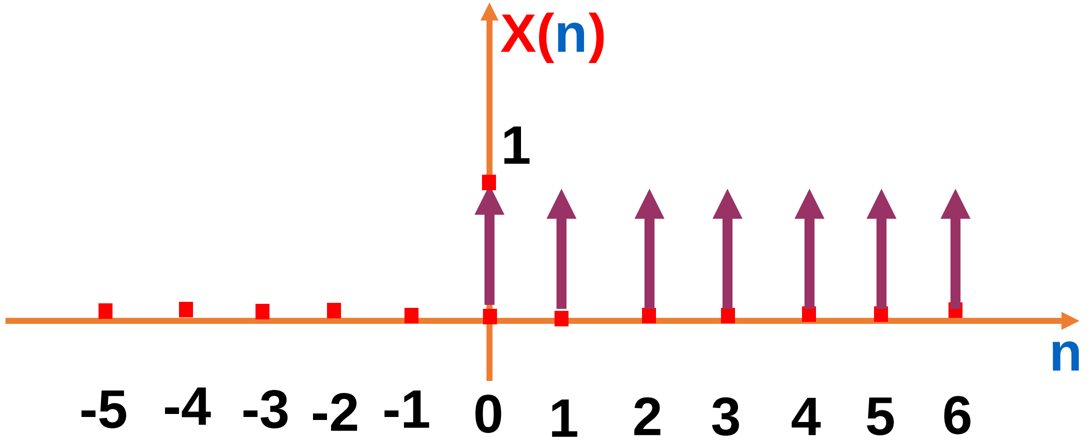
$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



DT Unit Step Sequence

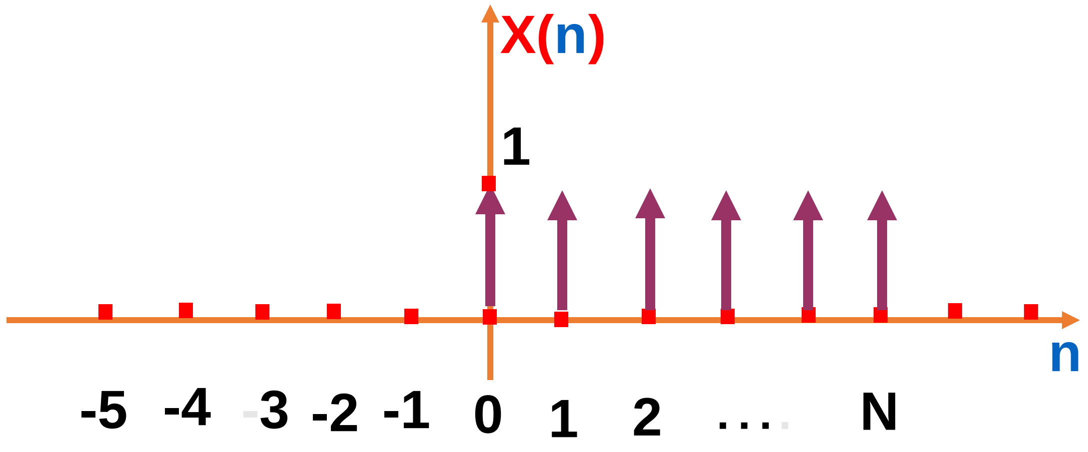
Unit Step (unit sample sequence) is described by:

$$U(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



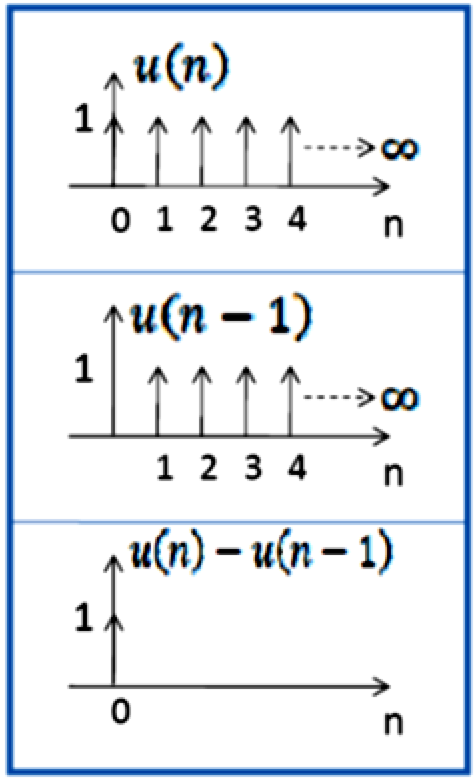
DT Rectangular Signal

$$x(n)=\begin{cases}1 & 0\le n\le N\\ 0 & otherwise\end{cases}$$

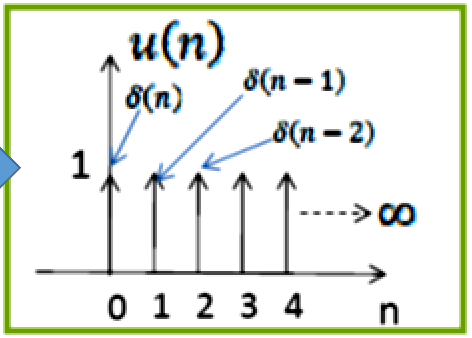


Relationship between Unit impulse and unit step

$$\boxed{\delta(n) = u(n) - u(n-1)}$$



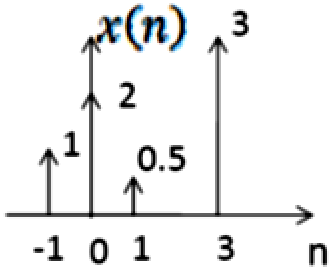
$$\boxed{u(n) = \sum_{k=0}^{\infty} \delta(n-k)}$$



Hence, the unit impulse signal can be used as a basic building block for the construction & representation of other signals.

e.g., $x(n) = \delta(n+1) + 2\,\delta(n) + 0.5\,\delta(n-1) + 3\,\delta(n-3)$

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$



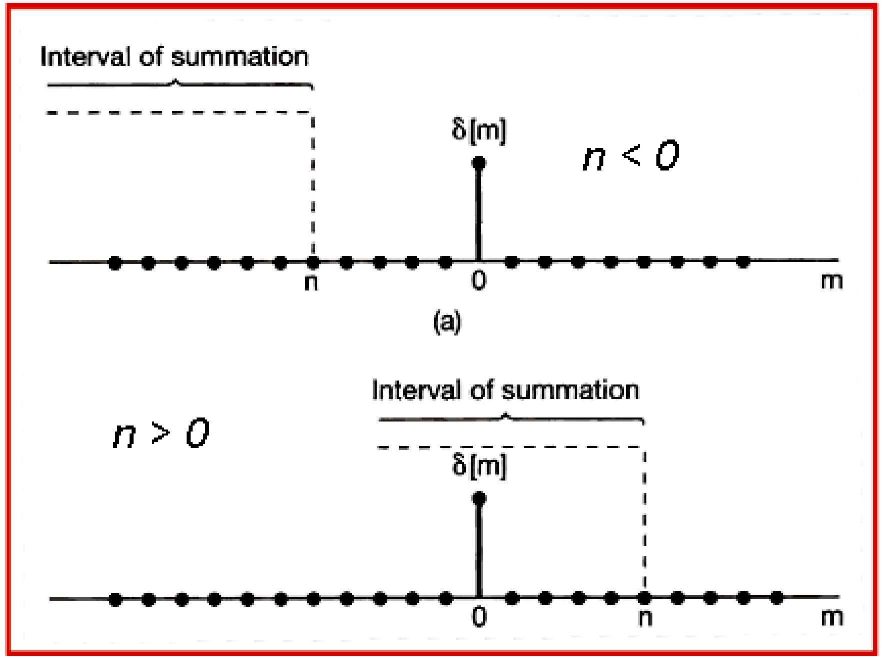
Relation Between Unit Impulse and Unit Step

The DT unit impulse is the **first difference** of the discrete time step

$$\delta[n] = u[n] - u[n-1]$$

Conversely, the DT unit step is the **running sum** of the unit sample.

$$u[n] = \sum_{m=-\infty}^{n} \delta[m]$$



As follows :

$$u[0] = \dots.. + \delta[-2] + \delta[-1] + \delta[0]$$
$$= \dots.. + 0 + 0 + 1 = 1$$

$$u[1] = \dots.. + \delta[-2] + \delta[-1] + \delta[0] + \delta[1]$$
$$= \dots\dots.. + 0 + 0 + 1 + 0 = 1$$

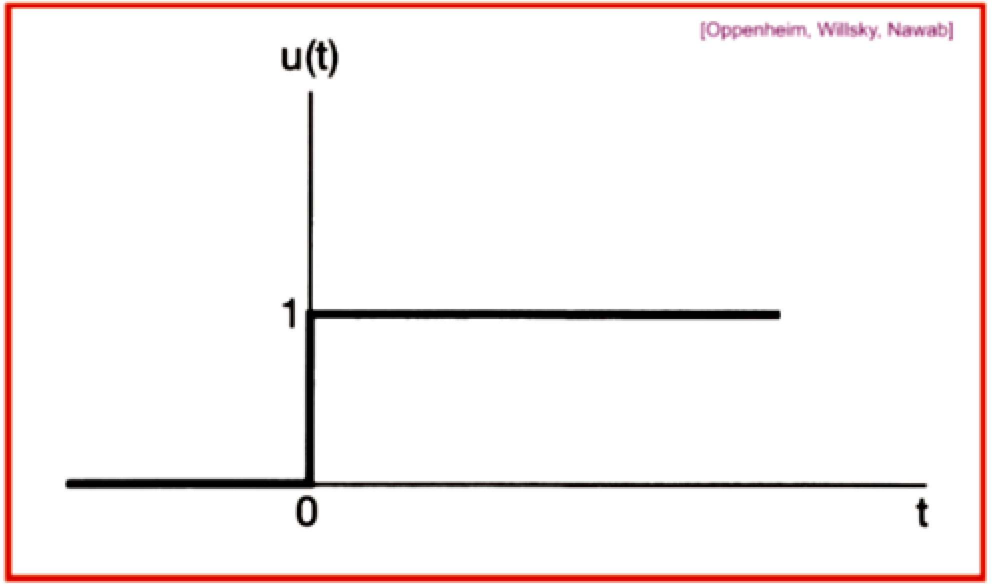
$$u[2] = \dots.. + \delta[-2] + \delta[-1] + \delta[0] + \delta[1] + \delta[2] = 1$$

CT Unit Step Signal

The CT **unit step function**

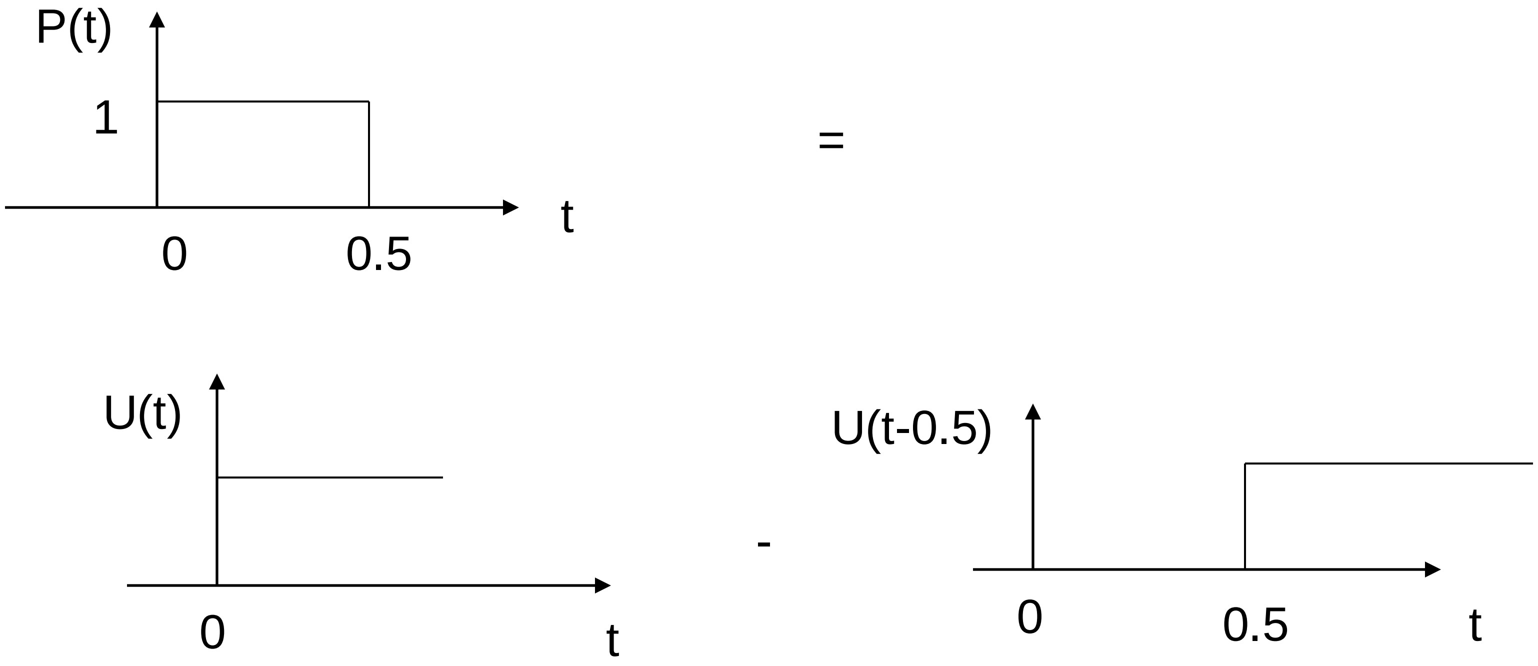
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

Note that the unit step function is discontinuous at $t = 0$.



Example of CT Unit Step Signal

- Example of this signal is a battery or dc source that is applied at t=0.
- Unit step signal is used to test how a system quickly responds to abrupt change in the input signal (unit step response).
- Unit step signal may be used to construct other discontinuous waveforms as follows:



Relation between Unit Impulse and Unit Step

The CT **unit impulse function** $\delta(t)$ is related to the unit step in a manner analogous to the relationship between the DT unit impulse and unit step. In particular $u(t)$ is the **running integral** of $\delta(t)$:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau$$

Same as Discrete

$$u[n] = \sum_{m=-\infty}^{n} \delta[m]$$

It can be thought, that unit impulse function is the first derivative of the CT unit step:

$$\delta(t) = \frac{du(t)}{dt}$$

Formal difficulty: $u(t)$ is not differentiable in the conventional sense because of its discontinuity at $t = 0$.

Continuous Approximation of Unit Impulse

$$U_\Delta(t) = \frac{1}{\Delta}t \qquad 0<t<\Delta$$
$$= 1 \qquad t > \Delta$$
$$=0 \qquad \text{otherwise}$$

Approximation of unit step function with $u_\triangle(t)$.

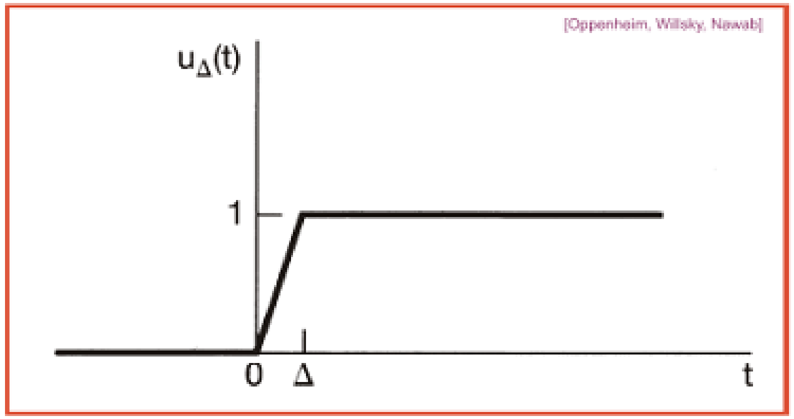
$$u(t) = \lim_{\triangle \to 0} u_\triangle(t)$$

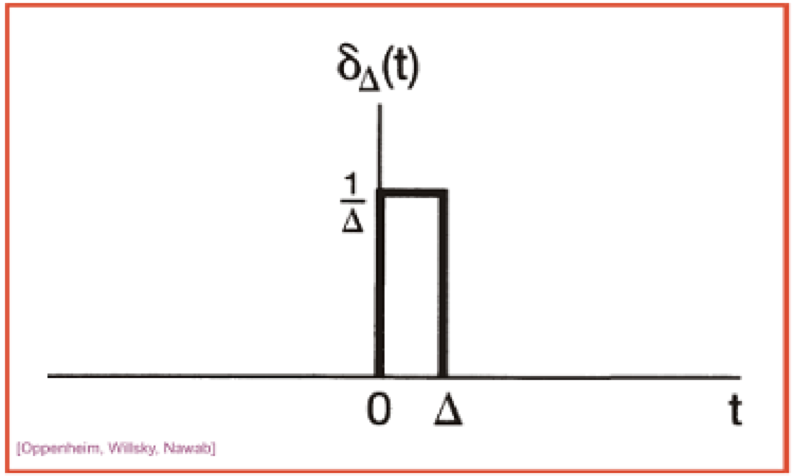
Let us consider the derivative

$$\delta_\triangle(t) = \frac{du_\triangle(t)}{dt}$$

The product of duration $\triangle$ and magnitude $1/\triangle$ of the $\delta_\triangle(t)$ function is always 1. Limit: definition of unit impulse function

$$\delta(t) \triangleq \lim_{\triangle \to 0} \delta_\triangle(t)$$





CT Unit Impulse Signal

$$\delta(t) = 0 \quad for \;\; t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

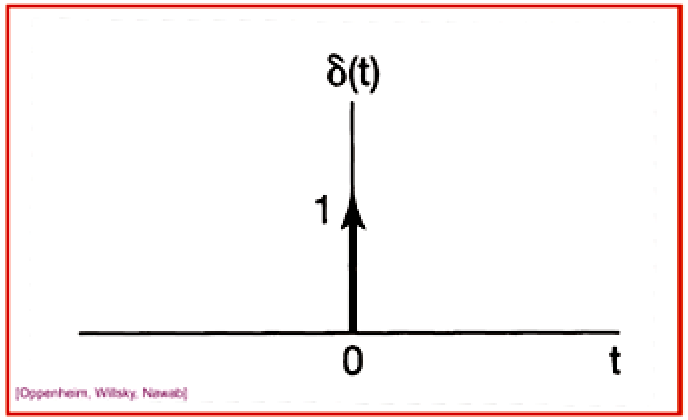
More generally, a scaled impulse $k\delta(t)$

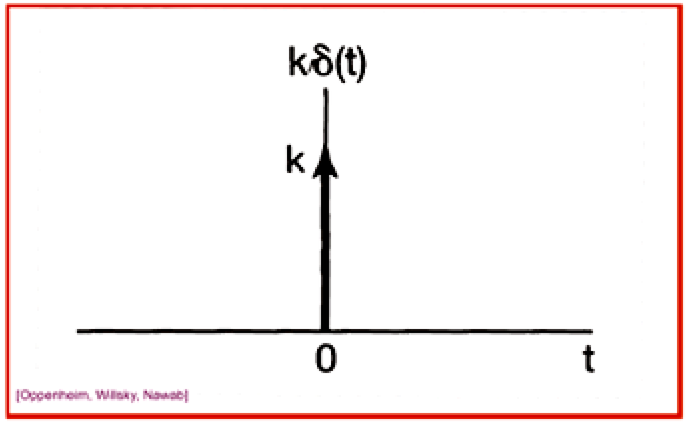
will have an area k,and thus

$$\int_{-\infty}^{t} k\delta(\tau)\, d\tau$$

$$= \; ku(t)$$

Symbol k next to arrow and the arrows hight are related to the impulse area.

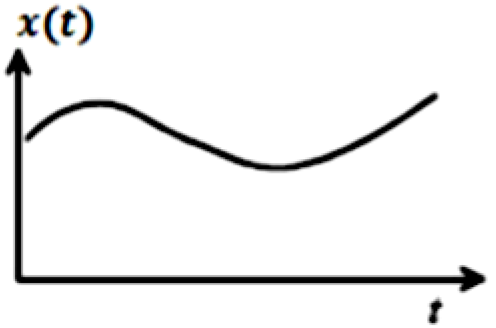




Application of CT Unit Impulse

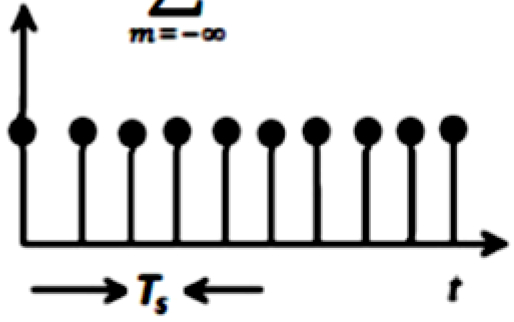
Unit area

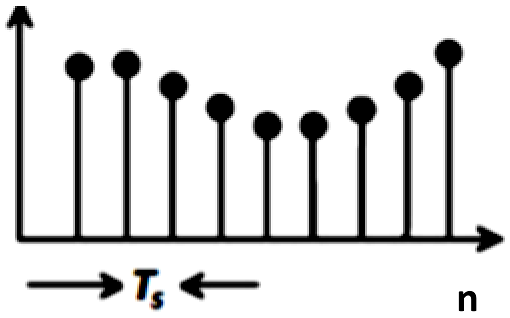
$$\int_{-\infty}^{\infty} \delta(\tau)d\tau = 1$$



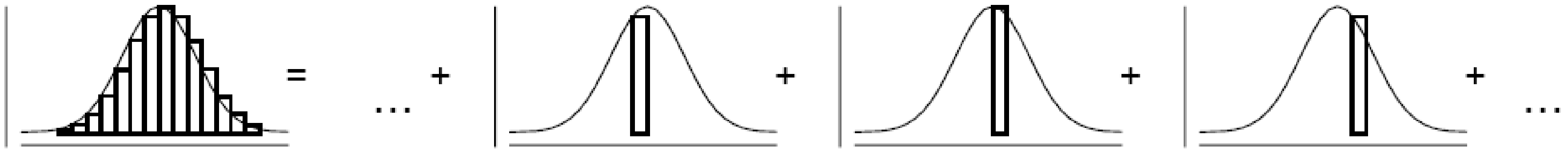
Unit Impulse used in sampling
(i.e to convert continuous time Signals to discrete time signals)

$$p(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT_s)$$





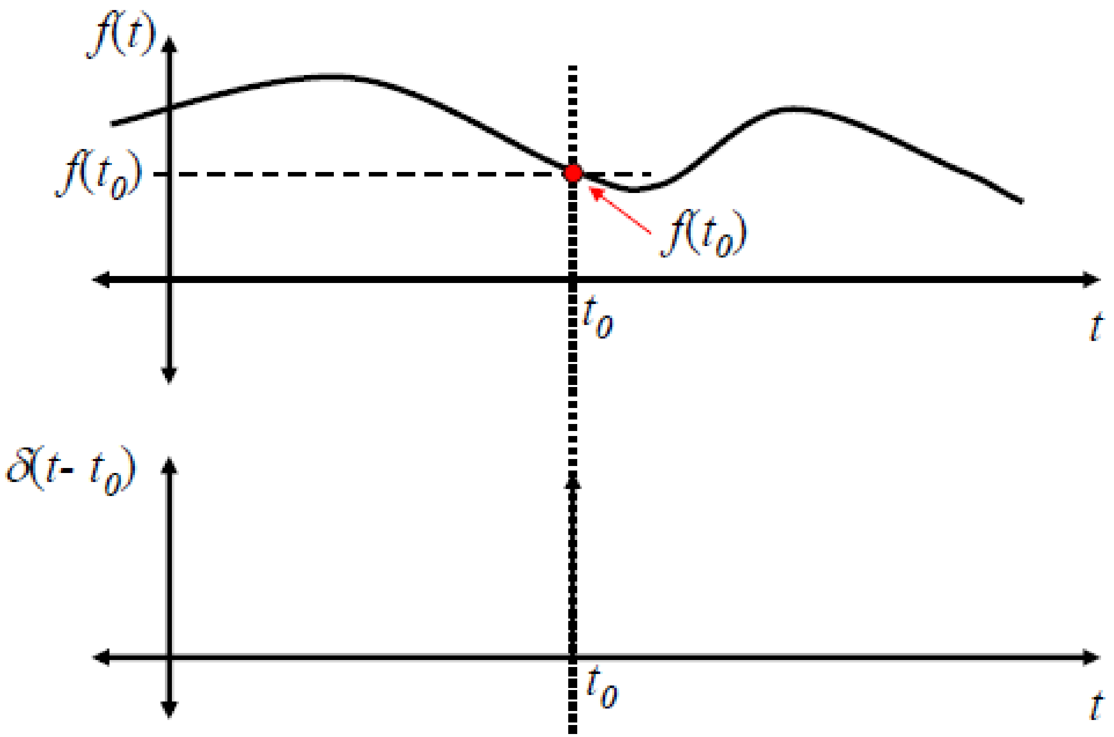
Representing CT Signals with Impulses



- **Conclusion:** we can represent any signal as an infinite sum of shifted and scaled unit impulses

The shifting Property is the most important property of $\delta(t)$:

$$\boxed{\int_{t_0-\varepsilon}^{t_0+\varepsilon} f(t)\delta(t-t_0)dt = f(t_0) \quad \forall \varepsilon > 0}$$



Integrating the product of $f(t)$ and $\delta(t-t_o)$ returns a single number… the value of $f(t)$ *at* the "location" of the shifted delta function

As long as the integral's limits surround the "location" of the delta… otherwise it returns zero

Examples:

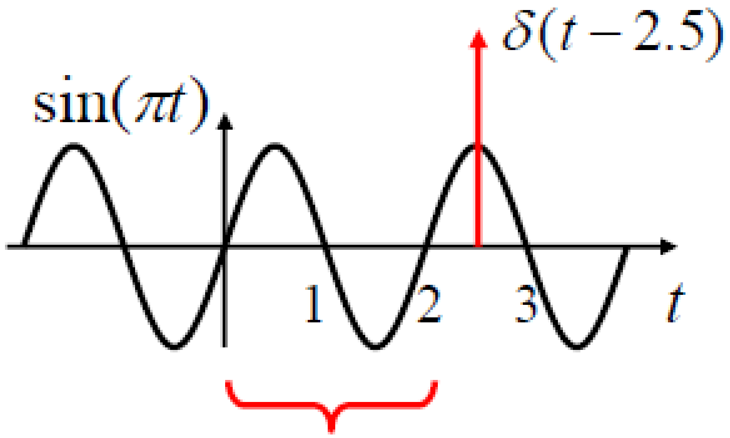
Find: $\displaystyle\int_0^2 \sin(\pi t)\delta(t-2.5)\,dt = ?$

Step 1: Find variable of integration: t

Step 2: Find the argument of $\delta(\bullet)$: $t-2.5$

Step 3: Find the value of the variable of integration that causes the argument of $\delta(\bullet)$ to go to zero: $t-2.5=0 \Rightarrow t=2.5$

Step 4: If value in Step 3 lies inside limits of integration… No! Otherwise… "return" zero…



$$\boxed{\int_0^2 \sin(\pi t)\delta(t-2.5)\,dt = 0}$$

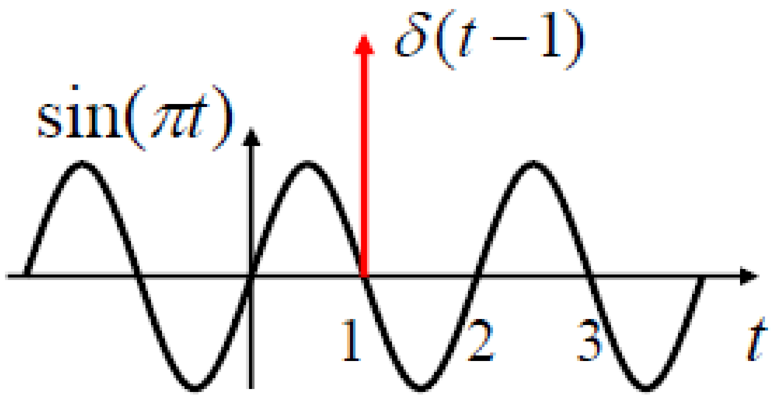
Range of Integration Does NOT "include delta function"

Examples:

$$\int_{t_0-\varepsilon}^{t_0+\varepsilon} f(t)\delta(t-t_0)dt = f(t_0) \quad \forall \varepsilon > 0$$

Find::

$$\int_{-4}^{7} \sin(\tfrac{\pi t}{2})\delta(t-1)dt = ?$$



Step 1: t Step 2: $t-1$

Step 3: $t-1=0 \;\Rightarrow\; t=1$

Step 4: $t=1$ lies in $[-4,7]$ so evaluate… $\sin(\frac{\pi}{2}\times 1) = \sin(\frac{\pi}{2}) = 1$

$$\boxed{\int_{-4}^{7} \sin(\tfrac{\pi t}{2})\delta(t-1)dt = 1}$$

1

Thank You